AISHA EVERING

Software Engineer | Computational Biology & Bioinformatics

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PROFESSIONAL SUMMARY

Software engineer with 20+ years of experience building production applications and graduate training in computational life sciences. Experienced in applying Python and R to biological data analysis, including transcriptomics, phylogenetics, genomics, statistical analysis, and data processing. Combines full-stack software engineering expertise with hands-on computational biology research to develop reliable tools, workflows, and analysis for complex biological data.

EDUCATION

ARIZONA STATE UNIVERSITY
Master of Science, Computational Life Sciences
October 2026

UNIVERSITY OF ADVANCING TECHNOLOGY
Bachelor of Science, Software Engineering
Associate of Science, Application Development

TEXAS STATE TECHNICAL COLLEGE
Associate of Science, Laser Electro-Optics

PROJECTS / PORTFOLIO

BIOTOOLS - BIOINFORMATICS COMMAND BUILDER

Designed and developed an interactive web application that translates user-selected alignment filtering criteria into reproducible `SAMtools view` commands, including SAM flag handling, quality thresholds, presets, and command decoding. Built with React and TypeScript and deployed as a production web application at bio-tools.dev.

TRANSCRIPTOMICS AND DIFFERENTIAL GENE EXPRESSION ANALYSIS

Analyzed RNA-seq count data in R using DESeq2, incorporating sample metadata, experimental design, dispersion estimation, and differential expression testing. Applied statistical significance and log2 fold-change thresholds, visualized gene expression patterns using volcano plots, and performed Gene Ontology enrichment analysis with clusterProfiler to investigate biological processes associated with differential expression.

DMC1 EVOLUTIONARY ANALYSIS IN METAMONADA

Conducted phylogenetic analysis of Dmc1 homologs across eukaryotes using EukPhylo, Taxus, IQ-TREE, FigTree, and HPC/Slurm workflows. Performed sequence curation, multiple sequence alignment, maximum-likelihood tree inference, and visualization. Investigated evolutionary relationships and potential gene duplication and loss across metamonad lineages. Contributed scientific interpretation, figures, and writing to a collaborative research manuscript and presented findings through a scientific poster.

ANCESTRY-RELATED DISPARITIES IN POLYGENIC RISK SCORE PERFORMANCE FOR TYPE 2 DIABETES

Investigated ancestry-related disparities in polygenic risk score performance for type 2 diabetes through a structured analysis of published studies, comparing predictive performance, population representation, and cross-ancestry transferability. Evaluated how training data composition and population differences affect model generalizability and identified persistent limitations in the equitable application of PRS across diverse populations.

BIOLOGICAL DATA STANDARDIZATION AND EVIDENCE MAPPING

Designed and implemented Python-based data processing pipelines to clean, normalize, and standardize heterogeneous research method data, improving data quality and enabling downstream analysis and evidence mapping.

SKILLS

BIOINFORMATICS & COMPUTATIONAL BIOLOGY

RNA-Seq Analysis, Differential Gene Expression, DESeq2, Gene Ontology Enrichment, clusterProfiler, Phylogenetic Analysis, Multiple Sequence Alignment, Sequence Curation, IQ-TREE, FigTree, EukPhylo, Taxus, SAMtools, FASTQ, BAM, VCF

PROGRAMMING & DATA ANALYSIS

Python, R, SQL/Relational Databases, Pandas, NumPy, Scikit-learn, ggplot2, Matplotlib, Plotly, Tableau, Jupyter Notebook

SOFTWARE ENGINEERING

TypeScript, JavaScript, React, C#, Java, Dart, Git, REST APIs, Full-Stack Development, Unit Testing, Agile

COMPUTING & MACHINE LEARNING

HPC, Slurm, Linux, Docker, Google Cloud, AWS (foundational), GitHub Actions (CI/CD), TensorFlow, PyTorch

EXPERIENCE

SOFTWARE ENGINEER, **WORKIVA** **Scottsdale, AZ**

Jan 2019 - Jan 2023

- Developed and enhanced features for a large-scale, customer-facing SaaS platform, translating product requirements into scalable, production-ready software.
- Built and maintained full-stack application functionality and data-processing workflows using Dart, JavaScript, and supporting technologies across front-end and backend systems.
- Designed solutions for transforming, integrating, and manipulating structured data to support application functionality, performance, and data-driven workflows.
- Collaborated with cross-functional Agile teams to design, test, troubleshoot, and deliver reliable software in a complex production environment.

SOFTWARE DEVELOPER, **US AIRWAYS/AMERICAN AIRLINES** **Phoenix, AZ**

Jan 2010 - Jan 2019

- Developed and maintained large-scale, customer-facing web applications for USAirways.com and AA.com, supporting high-volume customer traffic using C#, ASP.NET, Java, and related front-end and backend systems.
- Designed and implemented application features, backend services, and system integrations within complex enterprise environments, with an emphasis on performance, reliability, and maintainability.
- Contributed to large-scale system integration and migration initiatives following the US Airways-American Airlines merger, modernizing applications and improving data flow across platforms.
- Collaborated with geographically distributed, cross-functional teams throughout the software development lifecycle to design, troubleshoot, test, and deploy production systems.

CERTIFICATIONS

AWS Certified Cloud Practitioner - June 2024

DataCamp Data Scientist Associate; DataCamp Data Scientist - June-July 2024